The goal is to develop and deliver software to the customer rapidly without negatively affecting quality. Communication and collaboration between development and operations are, therefore, necessary.

The trend toward automation has helped DevOps culture bloom, because it codifies the cooperation between development and operations. You can do multiple deployments per day only if you automate the whole process. If you commit changes to the repository, the source code is automatically built and tested against your automated tests. If the build passes the tests, it's automatically installed in your testing environment, which triggers some acceptance tests. After those tests have passed, the change is propagated into production. But this isn't the end of the process: now you need to carefully monitor your system and analyze the logs in real time to ensure that the change was successful.

If your infrastructure is automated, you can spawn a new system for every change introduced to the code repository and run the acceptance tests isolated from other changes that were pushed to the repository at the same time. Whenever a change is made to the code, a new system is created (virtual machine, databases, networks, and so on) to run the change in isolation.

4.1.1 Why should you automate?

Why should you automate instead of using the graphical AWS Management Console? A script or a blueprint can be reused and will save you time in the long run. You can build new infrastructures quickly with ready-to-use modules from your former projects, or automate tasks that you will have to do regularly. Automating your infrastructure also enhances your software development process, for example, by using a deployment pipeline.